

Small-time Controllability of Control Systems on 3D Lie Groups

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Project title: Small-time controllability of control systems on 3D Lie groups

Project description: The goal of the project is to determine small-time controllability of affine control systems with one control on 3D unimodular Lie groups. By a parameter count one can check that all such control systems are equivalent to a finite number of model ones. After the classification is done, minimum controllability time will be estimated. This problem has some applications in quantum control, where the minimum controllability time is called quantum speed limit.

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1 Introduction

As motivation, the Schrodinger equation is

$$i\dot{\psi} = -\frac{\Delta}{2}\psi + \sum u_i(t)V_i(t)$$

In this project, we are studying the control system

$$\dot{g} = (A + uB)g, \quad G \subset GL(n) \tag{1.1}$$

where u is the control input. We are particularly interested in working with the case $\dim G = 3$, the unimodular Lie groups of dimension 3, which can be completely classified into five types of Lie groups: $SL(2)$, $SU(2)$, H^3 , $SE(2)$, and $SH(2)$.

This project involves two key goals

1. Classify systems of the type 1.1 on all unimodular Lie groups of dimension 3
2. Find all examples of this system 1.1 which are controllable in arbitrary small time

Todo: need to give more motivation.

2 Preliminaries

2.1 Groups

We first give the basic notions of groups, along with some examples.

Definition 2.1. A group G is a non-empty set with an operation $G \times G \rightarrow G$, $(a, b) \mapsto ab$ such that the following axioms hold:

1. $(ab)c = a(bc)$ (associativity)
2. $\exists e \in G$ such that $eg = ge$ for all $g \in G$ (identity)
3. $\forall g \in G, \exists d \in G$ such that $dg = gd = e$ (inverse)

A group is called commutative or Abelian if for all $g, h \in G$, we have $gh = hg$.

Example 2.2. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{F}, M_2(\mathbb{F}), \mathbb{Z}/n\mathbb{Z}$, all with the addition operation, are groups.

Example 2.3. Some more examples of groups

- Permutation group S_n
- Dihedral group D_8
- General linear group $GL(n, \mathbb{F})$

Definition 2.4. Let G be a group. A subset $H \subset G$ is called a subgroup if

1. $e \in H$
2. $a, b \in H \Rightarrow ab \in H$
3. $a \in H \Rightarrow a^{-1} \in H$

A subgroup H is a group in its own right.

Definition 2.5. Let G be a group. G is called cyclic if there is a $g \in G$ such that any element of G is a power of g , i.e. $G = \{g^n : n \in \mathbb{Z}\}$. This element g is called a generator of G .

Next, we introduce group actions on sets.

Definition 2.6. Let G be a group and let S be a non-empty set. G acts on S if we are given a function $G \times S \rightarrow S$, $(g, s) \mapsto gs$ such that

1. $es = s, \forall s \in S$
2. $(g_1g_2)s = g_1(g_2s), \forall g_1, g_2 \in G, s \in S$

Definition 2.7. We are given an action of G on S , and an element $s \in S$. The orbit of s is

$$\text{Orb}(s) = \{gs : g \in G\}$$

The stabiliser of s is

$$\text{Stab}(s) = \{g \in G : gs = s\}$$

Note that $\text{Orb}(s)$ is a subset of S , while $\text{Stab}(s)$ is a subset (in fact a subgroup) of G . Below are some examples.

Example 2.8. Let the group be $G = S_n$, let the set be $T = \{1, 2, \dots, n\}$. Then for $i \in T$, $\sigma i = \sigma(i)$, and $\text{Orb}(i) = T$.

Example 2.9. Let the group be $G = \mathbb{R}$, let the set be $S = \text{sphere in } \mathbb{R}^3 \text{ of radius 1 about the origin}$. $\mathbb{R}$ acts by rotating the sphere around the z -axis. The element $r \in \mathbb{R}$ rotates r radians. For $s \in S$, $\text{Orb}(s)$ is the altitude line on the sphere.

Example 2.10. Let G be a group, let $H \subset G$ be a subgroup. Here, we let H be the group and G be the set. H acts on G by $H \times G \rightarrow G$, $(h, g) \mapsto hg$. For $g \in G$, $\text{Orb}(g) = \{hg : h \in H\} = Hg = \text{right cosets of } H$.

Definition 2.11. A group action of G on X is called transitive if $\forall x, y \in X, \exists g \in G$ such that $gx = y$.

2.2 The Exponential Map

Let $M_n(\mathbb{F})$ be the space of $n \times n$ matrices with coefficients in the field $\mathbb{F}$ ($\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$). We consider it as a normed vector space with norm

$$\|X\| = \sup_{u \neq 0} \frac{|X(u)|}{|u|}$$

where $|u|$ is the usual norm in $\mathbb{R}^n$ or the hermitian norm in $\mathbb{C}^n$.

Definition 2.12. For $X \in M_n(\mathbb{F})$, we define

$$\exp(X) = \sum_{k=0}^{\infty} \frac{X^k}{k!}$$

Using the Cauchy-Schwartz inequality and the property $\|XY\| \leq \|X\|\|Y\|$, we can conclude that this series is normally convergent on compact sets.

Theorem 2.13. Let $X \in M_n(\mathbb{R})$ be a fixed matrix. Then the map $\exp : \mathbb{R} \rightarrow M_n(\mathbb{R})$ defined by

$$t \mapsto \exp(tX)$$

is analytic and is the unique solution to the differential equation

$$\frac{dY}{dt} = XY$$

with initial condition $Y(0) = \text{Id}$.

Proposition 2.14. The exponential map satisfies the following properties

1. For any $g \in GL(n, \mathbb{R})$ and $X \in M_n(\mathbb{R})$,

$$g(\exp(X))g^{-1} = \exp(gXg^{-1})$$

2. If $X, Y \in M_n(\mathbb{R})$ are two commuting matrices then

$$\exp(X + Y) = \exp(X) \exp(Y)$$

3. For any $X \in M_n(\mathbb{R})$

$$\exp(-X) = (\exp(X))^{-1}$$

Proposition 2.15 (Jacobi's formula). For any $X \in M_n(\mathbb{F})$ we have

$$\det(\exp X) = e^{\text{tr}(X)}$$

2.3 Lie Groups, Lie Algebras

Definition 2.16. A Lie group is a topological group which is a smooth manifold and such that the group operations, product $G \times G \rightarrow G$ and inverse map $G \rightarrow G$

$$(g, h) \mapsto gh \text{ and } g \mapsto g^{-1}$$

are smooth.

Proposition 2.17. The group of invertible matrices of rank n , $GL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det A \neq 0\}$, is a Lie group.

We will mostly be dealing with Lie groups which are submanifolds of $GL(n, \mathbb{R})$, which are called linear Lie groups.

Definition 2.18. Let G and H be Lie groups. A Lie group morphism from G to H is a smooth map $f : G \rightarrow H$ which is a group homomorphism.

A Lie group isomorphism between G and H is a smooth diffeomorphism $f : G \rightarrow H$ which is a group isomorphism. When $G = H$, we call f a Lie group automorphism.

Definition 2.19. A linear representation of G is a Lie group morphism $f : G \rightarrow GL(V)$ where V is a finite dimensional real vector space.

A matrix representation of G is a Lie group morphism $f : G \rightarrow GL(n, \mathbb{R})$ or $f : G \rightarrow GL(n, \mathbb{C})$ for some $n \in \mathbb{N}$.

A representation of G is either a linear representation or a matrix representation.

Next, we move on to the topic of Lie algebras.

Definition 2.20. A Lie algebra over a field $\mathbb{F}$ is a vector space $\mathfrak{g}$ over $\mathbb{F}$ equipped with a bilinear map (called a Lie bracket)

$$[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$$

such that

1. $[X, Y] = -[Y, X]$ (skew-symmetry)
2. $[X, [Y, Z]] + [Z, [X, Y]] + [Y, [Z, X]] = 0$ (Jacobi identity)

Definition 2.21. A Lie subalgebra of $\mathfrak{g}$ is a vector subspace $\mathfrak{h}$ such that for all $X, Y \in \mathfrak{h}$, $[X, Y] \in \mathfrak{h}$.

Example 2.22. The vector space $M_n(\mathbb{F})$ is a Lie algebra with the Lie bracket defined for all $X, Y \in M_n(\mathbb{F})$ by $[X, Y] = XY - YX$.

To any Lie group, we can associate a Lie algebra. It is the tangent space at the identity element of the group. First, in the case of linear Lie groups, we can develop this idea without knowledge of the tangent space to a manifold.

Definition 2.23. Let $G \subset GL(n, \mathbb{R})$ be a linear group. Define

$$Lie(G) = \{X \in M_n(\mathbb{R}) : \exp(tX) \in G \forall t \in \mathbb{R}\}$$

Proposition 2.24. Let $G \subset GL(n, \mathbb{R})$ be a linear group. Then $Lie(G) \subset M_n(\mathbb{R})$ is a Lie subalgebra.

Definition 2.25. Let G be a linear Lie group and let $\mathfrak{g}$ be its Lie algebra.

1. The adjoint representation of the Lie group is the homomorphism

$$Ad : G \rightarrow Aut(\mathfrak{g})$$

given by $Ad(g) = Ad_g$.

2. The adjoint representation of the Lie algebra is the homomorphism

$$ad : \mathfrak{g} \rightarrow End(\mathfrak{g})$$

given by $ad(X) = ad_X$.

We refer to Ad_g as the adjoint map defined by g , and to Ad as the adjoint map. We say the derivation ad_X is the adjoint map defined by X . The following gives a relation between these maps.

Proposition 2.26. *Let G be a linear Lie group, let $\mathfrak{g}$ be its Lie algebra, and $X \in \mathfrak{g}$. Then*

$$\frac{d}{dt} Ad_{\exp(tX)}|_{t=0} = ad_X$$

Definition 2.27. *Let $\mathfrak{g}, \mathfrak{h}$ be Lie algebras over a field K . A Lie algebra morphism from $\mathfrak{g}$ to $\mathfrak{h}$ is a K -linear map $\phi : \mathfrak{g} \rightarrow \mathfrak{h}$ such that $\forall X, Y \in \mathfrak{g}, [\phi(X), \phi(Y)] = [\phi(X), \phi(Y)]$.*

A Lie algebra isomorphism is a bijective Lie algebra morphism (its inverse is also a Lie algebra morphism). When $\mathfrak{g} = \mathfrak{h}$, a Lie algebra isomorphism is called a Lie algebra automorphism.

Definition 2.28. *A Lie subalgebra of $\mathfrak{g}$ is a vector subspace $V \subset \mathfrak{g}$ such that $[X, Y] \in V$ for all $X, Y \in V$.*

Definition 2.29. *A representation of a Lie algebra $\mathfrak{g}$ is a Lie algebra morphism $\phi : \mathfrak{g} \rightarrow End(V)$ where V is a vector space.*

Definition 2.30. *Any vector space V can be endowed with the Lie bracket defined by $[X, Y] = 0$ for all $X, Y \in V$. Such a Lie algebra $(V, [\cdot, \cdot])$ is called abelian.*

2.4 Examples of Lie Groups, Lie Algebras

In the following two examples, $\mathbb{R}$ can be replaced by $\mathbb{C}$.

Example 2.31. *Some important Lie groups*

1. $GL(n, \mathbb{R}) = \{g \in M_n(\mathbb{R}) : \det g \neq 0\}$
2. $SL(n, \mathbb{R}) = \{g \in GL(n, \mathbb{R}) : \det g = 1\}$
3. $O(n, \mathbb{R}) = \{g \in GL(n, \mathbb{R}) : {}^t g g = I_n\}$
4. $SO(n, \mathbb{R}) = O(n, \mathbb{R}) \cap SL(n, \mathbb{R})$
5. $U(n) = \{g \in GL(n, \mathbb{C}) : {}^t \bar{g} g = I_n\}$
6. $SU(n) = U(n) \cap SL(n, \mathbb{C})$

Example 2.32. *Some important Lie algebras*

1. $\mathfrak{gl}(V)$ and $\mathfrak{gl}(n, K)$, respectively corresponding to $End_{\mathbb{K}}(V)$ and $M_n(K)$, where K is a field and V is a vector space over K
2. $\mathfrak{sl}(n, \mathbb{R}) = \{X \in \mathfrak{gl}(n, \mathbb{R}) : tr(X) = 0\}$

$$3. \mathfrak{so}(n, \mathbb{R}) = \{X \in \mathfrak{gl}(n, \mathbb{R}) : {}^t X + X = 0\}$$

$$4. \mathfrak{u}(n, \mathbb{R}) = \{X \in \mathfrak{gl}(n, \mathbb{C}) : {}^t \bar{X} + X = 0\}$$

$$5. \mathfrak{su}(n) = \mathfrak{u}(n) \cap \mathfrak{sl}(n, \mathbb{C})$$

Todo: need a more complete list, with classical Lie groups and algebras.

In particular for this project, we are focusing on the Lie groups $SL(2)$, $SU(2)$, H , $SE(2)$, and $SH(2)$.

$$1. SL(2, \mathbb{R}) = \{X \in M_2(\mathbb{R}) : \det A = 1\}.$$

$$2. SU(2) = \left\{ \begin{pmatrix} \alpha & -\bar{\beta} \\ \beta & \bar{\alpha} \end{pmatrix} : \alpha, \beta \in \mathbb{C}, |\alpha|^2 + |\beta|^2 = 1 \right\}.$$

$$3. H = \left\{ \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} : a, b, c \in \mathbb{R} \right\} \text{ is the Heisenberg group. More generally, we can write } H(A), \text{ where } A \text{ is a unitary ring.}$$

4. $SE(2)$ includes arbitrary combinations of translations and rotations in Euclidean space, but not reflections (to be distinguished with $E(n)$). These transformations preserve Euclidean distance.

5. $SH(2)$ is the same as $SE(2)$, except "rotation" is replaced by "hyperbolic rotation". These transformations preserve the Lorentz form.

2.5 Killing Form

A property of finite dimensional Lie algebras is that they come with a symmetric bilinear form, which will be important in the structure theory of Lie algebras.

Definition 2.33. Let $\mathfrak{g}$ be a finite dimensional Lie algebra. The Killing form of $\mathfrak{g}$ is the bilinear form B on $\mathfrak{g}$ defined by

$$B(X, Y) = \text{tr}(ad(X) \circ ad(Y)), \quad \forall X, Y \in \mathfrak{g}$$

Proposition 2.34. Let $\mathfrak{g}$ be a finite dimensional Lie algebra, and B its Killing form. Then B is symmetric and ad -invariant, i.e.

$$B(ad(X)Y, Z) + B(Y, ad(X)Z) = 0, \quad \forall X, Y, Z \in \mathfrak{g}$$

Perhaps a clearer way to express the symmetry property is to write $B(X, Y) = B(Y, X)$. Moreover, ad -invariance means $ad(\mathfrak{g})$ is included in the Lie algebra $\mathfrak{o}(B)$ of the Lie group $O(B)$, which is the group of linear isomorphisms of $\mathfrak{g}$ preserving B .

Proposition 2.35. Let G be a Lie group with Lie algebra $\mathfrak{g}$. For all $g \in G$, the adjoint map $Ad(g)$ preserves the Killing form of $\mathfrak{g}$, i.e.

$$B(Ad(g)X, Ad(g)Y) = B(X, Y), \quad \forall X, Y \in \mathfrak{g}$$

The proof of this proposition makes use of the property that

$$ad(Ad(g)X) = Ad(g) \circ ad(x) \circ Ad(g)^{-1}$$

as well as the property that the trace is invariant by conjugation in $GL(\mathfrak{g})$.

It is useful to note that the adjoint map $ad(X) : \mathfrak{g} \rightarrow \mathfrak{g}$ is given by

$$ad(X)Y = [X, Y]$$

I also include the definition of the centre of a Lie group or Lie algebra.

Definition 2.36. *The centre of G is*

$$Z(G) = \{g \in G : \forall x \in G \, gx = xg\}$$

Definition 2.37. *The centre of $\mathfrak{g}$ is*

$$\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} : \forall Y \in \mathfrak{g} \, [X, Y] = 0\}$$

Example 2.38. *Some examples of Killing forms:*

1. *The Killing form of $\mathfrak{gl}(n, \mathbb{R})$ is $B(X, Y) = 2n \operatorname{tr}(XY) - 2 \operatorname{tr}(X) \operatorname{tr}(Y)$.*
2. *The Killing form of $\mathfrak{sl}(n, \mathbb{R})$ is $B(X, Y) = 2n \operatorname{tr}(XY)$.*
3. *The Killing form of an Abelian Lie algebra vanishes.*

Todo: add computational examples of finding the Killing form.

See 3.2 for an example of finding the matrix of the Killing form.

2.6 Control Systems

2.6.1 Equivalent Control Systems

Let G be a Lie group, and let $\mathfrak{g}$ be its associated Lie algebra. Consider the control system 1.1 with $g \in G$ and $A, B \in \mathfrak{g}$. When classifying 1.1 on this particular Lie group, we want to simplify the pair (A, B) into an equivalent pair, say (A', B') . This equivalency is denoted $(A, B) \sim (A', B')$. What tools can we use in this classification?

The first is the adjoint action of a Lie group on its Lie algebra, $ad : G \times \mathfrak{g} \rightarrow \mathfrak{g}$. For each $g \in G$, it is the derivative of this action in the second argument at the neutral element of G . For a matrix Lie group G , it is simply the conjugation $ad_g X = gXg^{-1}$. After letting g act on both A and B , the Killing form $B(A, B)$ is invariant

$$B(gAg^{-1}, gBg^{-1}) = B(A, B)$$

Thus we can write the equivalency $(A, B) \sim (gAg^{-1}, gBg^{-1})$. In different Lie groups, the conjugation group action will have different interpretations, which will become clear in the following sections.

Second, we can rewrite the control u as $u = u' + \delta$ for any δ . This can be thought of as a perturbation to the control. Then,

$$A + uB = A + (u' + \delta)B = (A + \delta B) + u'B$$

and so we have the equivalence $(A, B) \sim ((A + \delta B), B)$.

Another way to modify the control is to scale the control by a constant, writing $u = \mu u'$. Then,

$$A + uB = A + (\mu u')B = A + u'(\mu B)$$

which shows the equivalence $(A, B) \sim (A, \mu B)$.

Finally, the fourth option is to perform time scaling, for example by choosing a new measurement of time. If we scale the time by a factor of $1/\lambda$, then there is the equivalence $(A, B) \sim (\lambda A, \lambda B)$. Clearly, $\lambda \neq 0$.

2.6.2 Small-Time Controllability

I will first introduce the notions of controllability for the system

$$\dot{x} = F(x, u(t)) \quad (2.1)$$

which may be nonlinear. Here, $x \in \mathbb{R}^n$ is the state of the system, and $u(\cdot) : [0, \infty) \rightarrow U \subset \mathbb{R}^m$ is the control. Assume that F is a smooth (i.e. C^∞) function of its arguments, and that $u(\cdot)$ is regular enough such that 2.1 with the initial condition $x(0) = x_0 \in \mathbb{R}^n$ has local uniqueness and existence of solutions.

Let $x(t, x_0, u)$ denote the solution at time t of 2.1 starting from x_0 at $t = 0$ and corresponding to a control function $u(\cdot)$.

Definition 2.39. *Reachable (attainable) sets starting from x_0 :*

- the reachable set from x_0 at time $\tau \geq 0$ is

$$\mathcal{A}(\tau, x_0) = \{x_1 \in \mathbb{R}^n : \exists u(\cdot) : [0, \tau] \rightarrow U, x(\tau, x_0, u) = x_1\};$$

- the reachable set from x_0 within time $\tau \geq 0$ is

$$\mathcal{A}(\leq \tau, x_0) = \cup_{t \in [0, \tau]} \mathcal{A}(t, x_0);$$

- the reachable set from x_0 is

$$\mathcal{A}(x_0) = \cup_{t \in [0, \infty)} \mathcal{A}(t, x_0)$$

Definition 2.40. *The system 2.1 is said to be*

- controllable if for every $x_0 \in \mathbb{R}^n$, $\mathcal{A}(x_0) = \mathbb{R}^n$;
- small-time controllable if for every $x_0 \in \mathbb{R}^n$ and $\tau > 0$, we have $\mathcal{A}(\leq \tau, x_0) = \mathbb{R}^n$;
- small-time locally controllable at x_0 if x_0 belongs to the interior of $\mathcal{A}(\leq \tau, x_0)$ for every $\tau > 0$.

Remark. We assumed that $x \in \mathbb{R}^n$. However, we could equally apply these definitions for $x \in M$, a smooth connected manifold.

The following theorem will be central in my work on small-time controllability in Section 4.

Theorem 2.41 (Rashevsky-Chow). *Let M be a smooth manifold and $X^1, \dots, X^m$ be m smooth vector fields on M . Assume that*

$$\text{Lie}\{X^1, \dots, X^m\}(x) = T_x M, \quad \forall x \in M$$

Then the control system

$$\dot{x} = \sum_{i=1}^m u_i X^i(x)$$

is locally controllable in any time at every point of M .

Remark. *If M is connected, then local controllability implies global controllability.*

The condition in the Rashevsky-Chow theorem is called the bracket generating condition, or Hörmander's condition.

3 Classifying Control Systems

3.1 Classifying Control Systems on $SU(2)$

We consider the system 1.1 with $g \in SU(2)$ and $A, B \in \mathfrak{su}(2)$. This is the most basic scenario because it is closely related to the usual Euclidean space $\mathbb{R}^3$.

The following matrices

$$u_1 = \frac{1}{2} \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}; \quad u_2 = \frac{1}{2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}; \quad u_3 = \frac{1}{2} \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$$

form a basis for $\mathfrak{su}(2)$. The structural constants are given by the following relations

$$[u_1, u_2] = u_3; \quad [u_2, u_3] = u_1; \quad [u_3, u_1] = u_2$$

Remark. The factor of $1/2$ is useful in that the Killing form $B(u_i, u_j) = \delta_{ij}$ instead of $4\delta_{ij}$, where δ_{ij} is the Kronecker delta function.

The conjugation group action is a group automorphism. We will use the following relation

$$\text{Aut}(\mathfrak{su}(2)) \cong SO(3)$$

(source?) That is, the automorphism group of the Lie algebra $\mathfrak{su}(2)$ is isomorphic to the 3D rotation group $SO(3)$. So the conjugation group action is a rotation in the usual $\mathbb{R}^3$ space.

Another way to see this is perhaps to observe the Killing form. For $A, B \in \mathfrak{su}(2)$, $B(A, A)$ and $B(B, B)$ represent the length of A and B respectively, and $B(A, B)$ represents the angle between A and B (all in the sense of the usual Euclidean geometry). The Killing form is preserved under conjugation. For any $A, B \in \mathfrak{su}(2)$ and $g \in SU(2)$,

$$B(gAg^{-1}, gAg^{-1}) = B(A, A)$$

$$B(gBg^{-1}, gBg^{-1}) = B(B, B)$$

$$B(gAg^{-1}, gBg^{-1}) = B(A, B)$$

This means lengths and angles are preserved under conjugation, which suggests that the conjugation group action can be understood as rotations in $\mathbb{R}^3$.

Begin by writing A and B as linear combinations of the basis elements.

$$A = a_1^0 u_1 + a_2^0 u_2 + a_3^0 u_3$$

$$B = b_1^0 u_1 + b_2^0 u_2 + b_3^0 u_3$$

We can construct a 3D space with u_1, u_2, u_3 as the axes, analogous to $\mathbb{R}^3$ space. First, applying a combination of conjugation group actions to both A and B , we can rotate such that B onto the u_3 axis. Then, we can apply another rotation about the u_3 axis which preserves B , while A is rotated onto the u_1 - u_3 plane.

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 u_1 + a_3^1 u_3, b_3^1 u_3) \\ &\sim (a_1^1 u_1, b_3^1 u_3) \\ &\sim (a_1^1 u_1, a_1^1 u_3) \\ &\sim (u_1, u_3) \end{aligned}$$

To make the above process more explicit, I will use the same notation as in 2.6.1. The second equivalence is obtained by shifting the control, choosing $\delta = -a_3^1/b_3^1$. The third equivalence is obtained by scaling the control, choosing $\mu = a_1^1/b_3^1$. The fourth equivalence is obtained by time scaling, choosing $\lambda = 1/a_1^1$.

This result means that all control systems on $SU(2)$ are equivalent to the pair (u_1, u_3) .

3.2 Classifying Control Systems on $SL(2)$

We consider the system 1.1 with $g \in SL(2, \mathbb{R})$ and $A, B \in \mathfrak{sl}(2)$. The following matrices

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}; \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

form a basis for $\mathfrak{sl}(2)$. The structural constants are given by the following relations

$$[H, E] = 2E; \quad [H, F] = -2F; \quad [E, F] = H$$

Recall that for matrices $u_i, u_j \in \mathfrak{sl}(2)$, the Killing form can be computed using the formula

$$B(u_i, u_j) = 4\text{tr}(u_i u_j)$$

Recall also that the matrix of the Killing form is the matrix K whose entries are $k_{ij} = B(u_i, u_j)$. Using the above basis elements, we will determine the Killing form matrix.

It can be calculated that $B(E, E) = 0$, $B(E, F) = 4$, $B(E, H) = 0$, $B(F, E) = 4$, $B(F, F) = 0$, $B(F, H) = 0$, $B(H, E) = 0$, $B(H, F) = 0$, $B(H, H) = 8$. Thus, the Killing form matrix is

$$K = \begin{pmatrix} 0 & 4 & 0 \\ 4 & 0 & 0 \\ 0 & 0 & 8 \end{pmatrix}$$

The eigenvalues of this matrix are -4, 4, and 8. There are 2 positive eigenvalues and 1 negative eigenvalue, so the Killing form has signature (2, 1).

Remark. *The above choice of basis is the most commonly seen one, and it is the one I will continue to use. However, we could have alternatively chosen the basis*

$$E = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}; \quad F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}; \quad H = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

which can make the geometry of transformations more apparent. Moreover, using this basis, the Killing form matrix is just a diagonal matrix

$$K = \begin{pmatrix} 8 & 0 & 0 \\ 0 & 8 & 0 \\ 0 & 0 & -8 \end{pmatrix}$$

In this case, computing the eigenvalues is easier, but the conclusion is the same.

From the metric signature (2, 1), we see that $SL(2)$ is a group consisting of transformations in Lorentzian space $\mathbb{R}^{2,1}$. In Euclidean space, we have the standard scalar product

$$\langle x, y \rangle = \sum_{i=1}^n x^i y^i, \quad x, y \in \mathbb{R}^n$$

while in Lorentzian space, we have the Minkowski scalar product

$$\langle \langle x, y \rangle \rangle = -x^0 y^0 + \sum_{i=1}^{n-1} x^i y^i, \quad x, y \in \mathbb{R}^{n-1,1}$$

Similarly, we can see how the norm is defined differently in the two spaces.

In Lorentzian space, vectors can be classified as elliptic, hyperbolic, or parabolic based on the sign of their norm, or equivalently, the sign of the Killing form.

Let $X \in \mathfrak{sl}(2)$. If $B(X, X) = 0$, then X is parabolic, and lies on the light cone. If $B(X, X) < 0$, then X is elliptic, and lies on the hyperboloid of two sheets inside the light cone. If $B(X, X) > 0$, then X is hyperbolic, and lies on the hyperboloid of one sheet outside the light cone. Some sources call these causality relations.

Lorentz transformations - including null rotations, rotations, and Lorentz boosts - act via the conjugation group action. It is important to note that there does not exist a Lorentz transformation from the upper or lower sheet of the hyperboloid of two sheets to the hyperboloid of one sheet. Furthermore, there does not exist a Lorentz transformation the upper to the lower sheet of the hyperboloid of two sheets. In short, a vector can only remain in the region of space where it originally belonged to.

The strategy for classification is to begin by using the sign of the Killing form of B to split the analysis into three major cases. In each of these cases, we can identify a representative. Since this representative and B lie in the same region of space, we can apply Lorentz transformations via conjugation to simplify B to this representative. It is indeed better to start with B than with A , because shifting the control, scaling the control, and time scaling all preserve the sign of the Killing form $B(B, B)$. This means that after simplifying B using Lorentz transformations, we can focus on simplifying A .

We first find suitable representatives. Notice that

$$B(E, E) = 0$$

$$B(H, H) = 8 > 0$$

$$B(E - F, E - F) = -8 < 0$$

So we choose E , H , $E - F$ as the representatives for the parabolic, hyperbolic, and elliptic cases, respectively.

If, in our convention, E is the representative for the upper half of the light cone, then $-E$ is the representative for the lower half of the light cone. The same can be said for the hyperboloid of two sheets, by taking $E - F$ and $F - E$. However, in either case, it is not necessary to discuss the upper and lower halves separately. One can be transformed to the other by simply scaling the control.

Begin by writing A and B as linear combinations of the basis elements.

$$A = a_1^0 E + a_2^0 F + a_3^0 H$$

$$B = b_1^0 E + b_2^0 F + b_3^0 H$$

Case 1: $B(B, B) > 0$, B is hyperbolic

By a Lorentz transformation, B can be simplified to a multiple of H .

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 = 0$, then

$$\begin{aligned} (A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\ &= (a_1^1 E + a_3^1 H, b_3^1 H) \\ &\sim (a_1^1 E, b_3^1 H) \\ &\sim (a_1^1 E, a_1^1 H) \\ &\sim (E, H) \end{aligned}$$

If $a_1^1 = 0$ and $a_2^1 \neq 0$, then

$$\begin{aligned}
(A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\
&= (a_2^1 F + a_3^1 H, b_3^1 H) \\
&\sim (a_2^1 F, b_3^1 H) \\
&\sim (a_2^1 F, a_2^1 H) \\
&\sim (F, H)
\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 \neq 0$, then we need to find elements in the stabiliser group of H ,

$$Stab(H) = \{g \in SL(2) : gHg^{-1} = H\}$$

in order to continue simplifying, while preserving H . For $g \in SL(2)$, g must take the form

$$g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad ad - bc = 1$$

Carrying out the computations shows that elements of the stabiliser must be of the form

$$g = \begin{pmatrix} a & 0 \\ 0 & d \end{pmatrix}, \quad ad = 1$$

This matrix appears to represent a Lorentz boost, which usually is given as

$$\begin{pmatrix} e^{\eta/2} & 0 \\ 0 & e^{-\eta/2} \end{pmatrix}$$

Since

$$\begin{aligned}
gEg^{-1} &= a^2 E \\
gFg^{-1} &= d^2 F \\
gHg^{-1} &= H
\end{aligned}$$

we have

$$\begin{aligned}
g(a_1^1 E + a_2^1 F + a_3^1 H)g^{-1} &= a_1^1 a^2 E + a_2^1 d^2 F + a_3^1 H \\
g(b_3^1 H)g^{-1} &= b_3^1 H
\end{aligned}$$

$$\begin{aligned}
(A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_3^1 H) \\
&\sim (a_1^1 a^2 E + a_2^1 d^2 F + a_3^1 H, b_3^1 H)
\end{aligned}$$

It is possible to choose a and d such that $a_1^1 a^2 = a_2^1 d^2 = c$. The precise choices for a and d can be found by using $a_1^1 a^2 = a_2^1 d^2$ along with $ad = 1$. It is clear that the solution will involve a fourth root, and solutions only exist in the current scenario, where $a_1^1 \neq 0$ and $a_2^1 \neq 0$.

$$\begin{aligned}
(A, B) &\sim (c(E + F), b_3^1 H) \\
&\sim (c(E + F), cH) \\
&\sim (E + F, H)
\end{aligned}$$

Taking note of the sign, the final result should be $(\pm E \pm F, H)$.

Case 2: $B(B, B) < 0$, B is elliptic

By a Lorentz transformation, B can be simplified to a multiple of $E - F$.

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 (E - F))\end{aligned}$$

To continue simplifying, we want to find elements $g \in SL(2)$ that are also elements of the stabiliser group

$$Stab(E - F) = \{g \in SL(2) : g(E - F)g^{-1} = E - F\}$$

For this to be true, g must be of the form

$$g = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}, \quad a^2 + b^2 = 1$$

This resembles a rotation matrix

$$\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

and so the Lorentz transformations in this case act as rotations in a plane. If we fix $E - F$ as one axis, then for a 3D space, we need to construct two more, and it is on the plane formed by these two axes that we will rotate. Consider

$$E + F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \text{ and } H = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

There are the relations

$$B(E - F, H) = 0$$

$$B(E - F, E + F) = 0$$

$$B(E + F, H) = 0$$

so $E - F$, $E + F$ and H are indeed orthogonal, and can form a set of coordinate axes. Finding these two remaining axes can be thought of as finding the orthogonal complement of B ,

$$B^\perp = \{X \in \mathfrak{sl}(2) : B(B, X) = 0\} = \text{span}\{E + F, H\}$$

The strategy is then to rewrite A and B in terms of this new set of coordinates, reduce A onto $B^\perp$ by control shifting, then rotating on $B^\perp$. The steps are as follows:

$$\begin{aligned}(A, B) &\sim (a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 (E - F)) \\ &\sim (a_1^2 (E + F) + a_3^2 H, b_1^1 (E - F)) \\ &\sim (a_1^3 (E + F), b_1^1 (E - F)) \\ &\sim (a_1^3 (E + F), a_1^3 (E - F)) \\ &\sim (2a_1^3 E, a_1^3 (E - F)) \\ &\sim (2a_1^3 E, 2a_1^3 (E - F)) \\ &\sim (E, E - F)\end{aligned}$$

In the above, you can check that

$$a_1^2 = \frac{a_1^1 + a_2^1}{2}, \quad a_2^2 = \frac{a_1^1 - a_2^1}{2}, \quad a_3^2 = a_3^1$$

Remark. We could have equivalently rotated to H , which would give $(H, E - F)$. It does not matter which one we rotate to. Indeed, $B(E + F, E + F) = 8 > 0$, $B(H, H) = 8 > 0$, so for $A = \alpha(E + F) + \beta H$, $B(A, A) > 0$, hence rotations are all occurring on the hyperboloid of one sheet.

Case 3: $B(B, B) = 0$, B is parabolic

By a Lorentz transformation, B can be reduced to a multiple of E .

$$\begin{aligned}(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^1 E + a_2^1 F + a_3^1 H, b_1^1 E)\end{aligned}$$

Then, we find elements $g \in SL(2)$ such that g is also an element of the stabiliser group of E

$$Stab(E) = \{g \in SL(2) : gEg^{-1} = E\}$$

Such elements are of the form

$$g = \begin{pmatrix} a & b \\ 0 & a \end{pmatrix}, \quad a^2 = 1$$

This matrix resembles the matrix for the null rotation

$$\begin{pmatrix} 1 & \alpha \\ 0 & 1 \end{pmatrix}$$

From the relations

$$\begin{aligned}gEg^{-1} &= a^2 E = E \\ gFg^{-1} &= -b^2 E + F + abH \\ gHg^{-1} &= -2abE + H\end{aligned}$$

we have

$$\begin{aligned}gAg^{-1} &= (a_1^1 - b^2 a_2^1 - 2aba_3^1)E + a_2^1 F + (aba_2^1 + a_3^1)H \\ gBg^{-1} &= b_1^1 E\end{aligned}$$

If $a_2^1 = 0$, then

$$\begin{aligned}(A, B) &\sim (a_3^1 H, b_1^1 E) \\ &\sim (a_3^1 H, a_3^1 E) \\ &\sim (H, E)\end{aligned}$$

If $a_2^1 \neq 0$, then choose $b = -a_3^1/aa_2^1$ to eliminate the H component in A , thus

$$\begin{aligned}(A, B) &\sim (a_2^1 F, b_1^1 E) \\ &\sim (a_2^1 F, a_2^1 E) \\ &\sim (F, E)\end{aligned}$$

3.3 Classifying Control Systems on H

We consider the system 1.1 with $g \in H$ and $A, B \in \mathfrak{h}$. The following matrices

$$X = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}; \quad Z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

form a basis for $\mathfrak{h}$. The structural constants are given by the following relations

$$[X, Y] = Z; \quad [X, Z] = 0; \quad [Y, Z] = 0$$

The Heisenberg group is different from the previous two scenarios in that for any $B \in \mathfrak{h}$, it can be verified that the Killing form $B(B, B) = 0$. So the Killing form is not so useful for classification here.

We start by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z$$

$$B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

There are two types of transformations which will be useful. The first are the inner automorphisms. For $M \in \mathfrak{h}$ and $g \in H$, inner automorphisms are mappings of the form $M \mapsto gMg^{-1}$. If we write M as

$$M = m_1 X + m_2 Y + m_3 Z = \begin{pmatrix} 0 & m_1 & m_3 \\ 0 & 0 & m_2 \\ 0 & 0 & 0 \end{pmatrix}$$

and g as

$$g = \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

then the inner automorphism can be computed as

$$\begin{aligned} gMg^{-1} &= \begin{pmatrix} 1 & a & c \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & m_1 & m_3 \\ 0 & 0 & m_2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & -a & ab - c \\ 0 & 1 & -b \\ 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 0 & m_1 & -bm_1 + am_2 + m_3 \\ 0 & 0 & m_2 \\ 0 & 0 & 0 \end{pmatrix} \\ &= m_1 X + m_2 Y + (-bm_1 + am_2 + m_3)Z \end{aligned}$$

Notice that the inner automorphism preserves the X and Y components and only shifts the Z component. Applying this transformation to A and B , we obtain

$$gAg^{-1} = a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z$$

$$gBg^{-1} = b_1^0 X + b_2^0 Y + (-bb_1^0 + ab_2^0 + b_3^0)Z$$

The second type of transformations are the symplectic maps. The symplectic map uses a matrix $T \in Sp(2, \mathbb{R})$ to act by left matrix multiplication on the (X, Y) vector in M , and preserves the Z component. In symbols, we can write the mapping

$$((X, Y), Z) \mapsto (T(X, Y), Z)$$

Note that in the $n = 2$ case, $Sp(2, \mathbb{R}) = SL(2, \mathbb{R})$. So when $T \in Sp(2, \mathbb{R})$, we can take T to be the rotation matrix

$$T = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Doing so allows us to rotate in the (X, Y) plane.

The result of applying the inner automorphism to A and B - X and Y preserved with Z shifted - suggests that we should consider the centre of the Lie algebra $\mathfrak{h}$. By applying the definition of the centre of a Lie algebra, I found

$$\mathfrak{z}(\mathfrak{h}) = \left\{ \begin{pmatrix} 0 & 0 & c \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} : c \in \mathbb{R} \right\} = \{cZ : c \in \mathbb{R}\}$$

We try to simplify B first, and to do so, we analyze two cases: either B belongs to the centre, or B does not belong to the centre.

Case 1: $B \in \mathfrak{z}(\mathfrak{h})$

Here, $b_3^0 \neq 0$, while $b_1^0 X + b_2^0 Y = 0$, i.e. $b_1^0 = b_2^0 = 0$. Then,

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_3^0 Z) \\ &\sim (a_1^0 X + a_2^0 Y, b_3^0 Z) \end{aligned}$$

If $(a_1^0, a_2^0) = (0, 0)$, then

$$\begin{aligned} (A, B) &\sim (0, b_3^0 Z) \\ &\sim (0, Z) \end{aligned}$$

If $(a_1^0, a_2^0) \neq (0, 0)$, then using the rotation matrix T simultaneously on A and B , we can rotate $a_1^0 X + a_2^0 Y$ on to either X or Y , while $b_3^0 Z$ remains unchanged. I choose to rotate on to X , and so

$$\begin{aligned} (A, B) &\sim (a_1^1 X, b_3^0 Z) \\ &\sim (a_1^1 X, a_1^1 Z) \\ &\sim (X, Z) \end{aligned}$$

Case 2: $B \notin \mathfrak{z}(\mathfrak{h})$

Here, $b_1^0 X + b_2^0 Y \neq 0$, so $(b_1^0, b_2^0) \neq (0, 0)$. b_3^0 may or may not be 0. In the inner automorphism on B ,

$$gBg^{-1} = b_1^0 X + b_2^0 Y + (-bb_1^0 + ab_2^0 + b_3^0)Z$$

since $(b_1^0, b_2^0) \neq (0, 0)$, we can choose the entries $a, b \in \mathbb{R}$ of the matrix g such that $-bb_1^0 + ab_2^0 + b_3^0 = 0$, and so

$$gBg^{-1} = b_1^0 X + b_2^0 Y$$

This can further be rotated on to X by applying the symplectic map to A and B . We have the following

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= (a_1^0 X + a_2^0 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^0 X + b_2^0 Y) \\ &\sim (a_1^1 X + a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &\sim (a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \end{aligned}$$

If $a_2^1 = 0$, then

$$\begin{aligned}(A, B) &= ((-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &= ((-ba_1^0 + aa_2^0 + a_3^0)Z, (-ba_1^0 + aa_2^0 + a_3^0)X) \\ &= (Z, X)\end{aligned}$$

If $a_2^1 \neq 0$, then we need another inner automorphism that is at the same time an element of the stabiliser of X . Take another $h \in H$ of the form

$$h = \begin{pmatrix} 1 & a' & c' \\ 0 & 1 & b' \\ 0 & 0 & 1 \end{pmatrix}$$

For h to be an element of $Stab(X)$, we require $hXh^{-1} = X$. Computing this gives

$$\begin{pmatrix} 0 & 1 & -b' \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

thus we need $b' = 0$ to ensure $hXh^{-1} = X$. Moreover,

$$hYh^{-1} = \begin{pmatrix} 0 & 0 & a' \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = Y + a'Z$$

$$hZh^{-1} = Z$$

We now apply this inner automorphism.

$$\begin{aligned}(A, B) &\sim (a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &\sim (h(a_2^1 Y + (-ba_1^0 + aa_2^0 + a_3^0)Z)h^{-1}, h(b_1^1 X)h^{-1}) \\ &= (a_2^1 (Y + a'Z) + (-ba_1^0 + aa_2^0 + a_3^0)Z, b_1^1 X) \\ &= (a_2^1 Y + [a_2^1 a' + (-ba_1^0 + aa_2^0 + a_3^0)]Z, b_1^1 X)\end{aligned}$$

Since $a_2^1 \neq 0$, we can choose the entries of h to be $a' = -(-ba_1^0 + aa_2^0 + a_3^0)/a_2^1$, $b' = 0$, $c' \in \mathbb{R}$ so that $a_2^1 a' + (-ba_1^0 + aa_2^0 + a_3^0) = 0$. Proceeding with this,

$$\begin{aligned}(A, B) &\sim (a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, a_2^1 X) \\ &\sim (Y, X)\end{aligned}$$

3.4 Classifying Control Systems on $SE(2)$

The $SE(2)$ group is defined as follows

$$SE(2) = \left\{ \begin{pmatrix} \cos \theta & -\sin \theta & x \\ \sin \theta & \cos \theta & y \\ 0 & 0 & 1 \end{pmatrix} : \theta, x, y \in \mathbb{R} \right\}$$

I will derive the corresponding Lie algebra here, demonstrating a general method for doing so. To obtain the identity element of $SE(2)$, we need $\theta = 0$, $x = 0$, and $y = 0$. Parameterize $\theta(t)$, $x(t)$, $y(t)$ with $t \geq 0$ such that $\theta(0) = 0$, $x(0) = 0$, $y(0) = 0$. That is, these curves pass through the identity at $t = 0$. Then, we differentiate an element of $SE(2)$ at $t = 0$.

$$\begin{aligned} \left. \frac{d}{dt} \begin{pmatrix} \cos \theta(t) & -\sin \theta(t) & x(t) \\ \sin \theta(t) & \cos \theta(t) & y(t) \\ 0 & 0 & 1 \end{pmatrix} \right|_{t=0} &= \begin{pmatrix} -\sin \theta(t) \dot{\theta}(t) & -\cos \theta(t) \dot{\theta}(t) & \dot{x}(t) \\ \cos \theta(t) \dot{\theta}(t) & -\sin \theta(t) \dot{\theta}(t) & \dot{y}(t) \\ 0 & 0 & 0 \end{pmatrix} \Big|_{t=0} \\ &= \begin{pmatrix} 0 & -\dot{\theta}(0) & \dot{x}(0) \\ \dot{\theta}(0) & 0 & \dot{y}(0) \\ 0 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 0 & -c & a \\ c & 0 & a \\ 0 & 0 & 0 \end{pmatrix} \end{aligned}$$

Thus, the Lie algebra corresponding to $SE(2)$ is

$$\mathfrak{se}(2) = \left\{ \begin{pmatrix} 0 & -c & a \\ c & 0 & b \\ 0 & 0 & 0 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$$

The following matrices

$$X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}; \quad Z = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

form a basis for $\mathfrak{se}(2)$. These basis elements satisfy

$$[X, Y] = 0; \quad [Y, Z] = X; \quad [X, Z] = -Y.$$

Begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z$$

$$B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

For a matrix $g \in SE(2)$, we have

$$g = \begin{pmatrix} \cos \theta & -\sin \theta & x \\ \sin \theta & \cos \theta & y \\ 0 & 0 & 1 \end{pmatrix}, \quad g^{-1} = \begin{pmatrix} \cos \theta & \sin \theta & -x \cos \theta - y \sin \theta \\ -\sin \theta & \cos \theta & x \sin \theta - y \cos \theta \\ 0 & 0 & 1 \end{pmatrix}$$

where $\theta, x, y \in \mathbb{R}$. The conjugation group action on the basis elements gives

$$gXg^{-1} = \begin{pmatrix} 0 & 0 & \cos \theta \\ 0 & 0 & \sin \theta \\ 0 & 0 & 0 \end{pmatrix} = \cos \theta X + \sin \theta Y$$

$$gYg^{-1} = \begin{pmatrix} 0 & 0 & -\sin \theta \\ 0 & 0 & \cos \theta \\ 0 & 0 & 0 \end{pmatrix} = -\sin \theta X + \cos \theta Y$$

$$gZg^{-1} = \begin{pmatrix} 0 & -1 & y \\ 1 & 0 & -x \\ 0 & 0 & 0 \end{pmatrix} = yX - xY + Z$$

Either making use of the decomposition into basis elements or by direct calculation, we have

$$gAg^{-1} = (a_1^0 \cos \theta - a_2^0 \sin \theta + a_3^0 y)X + (a_1^0 \sin \theta + a_2^0 \cos \theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0 \cos \theta - b_2^0 \sin \theta + b_3^0 y)X + (b_1^0 \sin \theta + b_2^0 \cos \theta - b_3^0 x)Y + b_3^0 Z$$

Indeed, the transformations here are combinations of rotations and translations in Euclidean space, where rotations are determined by θ , and translations are determined by x and y .

There are two cases to consider: when $B \in \text{Span}(X, Y)$, and when $B \notin \text{Span}(X, Y)$.

Case 1: $B \in \text{Span}(X, Y)$

In this case, $B = b_1^0 X + b_2^0 Y$, with $(b_1^0, b_2^0) \neq 0$ and $b_3^0 = 0$. Since b_3^0 , we can disregard x and y , and choose θ so that gBg^{-1} rotates B in the (X, Y) plane on to X , then simplify.

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= ((a_1^0 \cos \theta - a_2^0 \sin \theta + a_3^0 y)X + (a_1^0 \sin \theta + a_2^0 \cos \theta - a_3^0 x)Y + a_3^0 Z, b_1^1 X) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X) \\ &\sim (a_2^1 Y + a_3^0 Z, b_1^1 X) \end{aligned}$$

If $a_3^0 = 0$, then assuming $a_2^1 \neq 0$,

$$\begin{aligned} (A, B) &\sim (a_2^1 Y, b_1^1 X) \\ &\sim (a_2^1 Y, a_2^0 X) \\ &\sim (Y, X) \end{aligned}$$

If $a_3^0 \neq 0$, then we need to find the elements of the stabiliser group of X , so that applying gAg^{-1} and gBg^{-1} preserves $b_1^1 X$. The stabiliser is

$$\text{Stab}(X) = \{g \in SE(2) : gXg^{-1} = X\}$$

From the result of gXg^{-1} computed above, it is clear that to preserve $b_1^1 X$, we just need $\theta = 0$. Moreover, noting the result of gZg^{-1} , we choose $x = a_3^1/a_3^0$ and $y = 0$ to eliminate the Y component and prevent X from reappearing in A . Thus,

$$\begin{aligned} (A, B) &\sim (a_3^0 Z, b_1^1 X) \\ &\sim (a_3^0 Z, a_3^0 X) \\ &\sim (Z, X) \end{aligned}$$

Case 2: $B \notin \text{Span}(X, Y)$

Here, $b_3^0 \neq 0$. When applying gAg^{-1} and gBg^{-1} , we can choose $x = 0$ and $y = 0$ so that we are just rotating in the (X, Y) plane.

$$\begin{aligned} (A, B) &\sim (gAg^{-1}, gBg^{-1}) \\ &= ((a_1^0 \cos \theta - a_2^0 \sin \theta + a_3^0 y)X + (a_1^0 \sin \theta + a_2^0 \cos \theta - a_3^0 x)Y + a_3^0 Z, b_1^1 X + b_3^0 Z) \\ &= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X + b_3^0 Z) \end{aligned}$$

If $b_1^1 = 0$, i.e. $B \in \text{Span}(Z)$, then

$$\begin{aligned}(A, B) &\sim (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^1 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

Take another $g \in SE(2)$ and choose $x = 0$, $y = 0$, and $\theta \in \mathbb{R}$ to perform a rotation on to X , while Z is preserved.

$$\begin{aligned}(A, B) &\sim (a_1^2 X, b_3^0 Z) \\ &\sim (a_1^2 X, a_1^2 Z) \\ &\sim (X, Z)\end{aligned}$$

If $b_1^1 \neq 0$, we take another $g \in SE(2)$ with $\theta = 0$ so that there is no rotation, only translation. We compute

$$\begin{aligned}g(a_1^1 X + a_2^1 Y + a_3^0 Z)g^{-1} &= (a_1^1 + a_3^0 y)X + (a_2^1 - a_3^0 x)Y + a_3^0 Z \\ g(b_1^1 X + b_3^0 Z)g^{-1} &= (b_1^1 + b_3^0 y)X - b_3^0 x Y + b_3^0 Z\end{aligned}$$

Choosing $x = 0$, $y = -b_1^1/b_3^0$ gives

$$\begin{aligned}(A, B) &\sim (a_1^2 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\ &\sim (a_1^2 X + a_2^1 Y, b_3^0 Z)\end{aligned}$$

Finally, take another $g \in SE(2)$ and set $x = 0$, $y = 0$. $\theta \in \mathbb{R}$ can be chosen such that $a_1^2 X + a_2^1 Y$ is rotated on to X .

$$\begin{aligned}(A, B) &\sim (g(a_1^2 X + a_2^1 Y)g^{-1}, g(b_3^0 Z)g^{-1}) \\ &= (a_1^3 X, b_3^0 Z) \\ &\sim (a_1^3 X, a_1^3 Z) \\ &\sim (X, Z)\end{aligned}$$

3.5 Classifying Control Systems on $SH(2)$

The $SH(2)$ group is similar to $SE(2)$. While $SE(2)$ is translations and rotations in Euclidean space preserving Euclidean distance, $SH(2)$ is translations and hyperbolic rotations preserving the Lorentz form. It is given as follows

$$SH(2) = \left\{ \begin{pmatrix} \cosh \theta & \sinh \theta & x \\ \sinh \theta & \cosh \theta & y \\ 0 & 0 & 1 \end{pmatrix} : \theta, x, y \in \mathbb{R} \right\}$$

Recall that $\sinh$ and $\cosh$ are the hyperbolic trigonometric functions defined by

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2}$$

with the relation

$$\cosh^2 x - \sinh^2 x = 1$$

In the following, it will be useful to remember that $\sinh 0 = 0$ and $\cosh 0 = 1$.

In a similar fashion to the $SE(2)$ case, the Lie algebra corresponding to the $SH(2)$ group can be derived to be

$$\mathfrak{sh}(2) = \left\{ \begin{pmatrix} 0 & c & a \\ c & 0 & b \\ 0 & 0 & 0 \end{pmatrix} : a, b, c \in \mathbb{R} \right\}$$

The matrices

$$X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}; \quad Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}; \quad Z = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

form a basis for $\mathfrak{sh}(2)$. These basis elements satisfy

$$[X, Y] = 0; \quad [Y, Z] = -X; \quad [X, Z] = -Y.$$

Again, begin by writing A and B as linear combinations of the basis elements

$$A = a_1^0 X + a_2^0 Y + a_3^0 Z$$

$$B = b_1^0 X + b_2^0 Y + b_3^0 Z$$

For a matrix $g \in SH(2)$, we have

$$g = \begin{pmatrix} \cosh \theta & \sinh \theta & x \\ \sinh \theta & \cosh \theta & y \\ 0 & 0 & 1 \end{pmatrix}, \quad g^{-1} = \begin{pmatrix} \cosh \theta & -\sinh \theta & -x \cosh \theta + y \sinh \theta \\ -\sinh \theta & \cosh \theta & x \sinh \theta - y \cosh \theta \\ 0 & 0 & 1 \end{pmatrix}$$

where $\theta, x, y \in \mathbb{R}$. The conjugation group action on the basis elements gives

$$gXg^{-1} = \begin{pmatrix} 0 & 0 & \cosh \theta \\ 0 & 0 & \sinh \theta \\ 0 & 0 & 0 \end{pmatrix} = \cosh \theta X + \sinh \theta Y$$

$$gYg^{-1} = \begin{pmatrix} 0 & 0 & \sinh \theta \\ 0 & 0 & \cosh \theta \\ 0 & 0 & 0 \end{pmatrix} = \sinh \theta X + \cosh \theta Y$$

$$gZg^{-1} = \begin{pmatrix} 0 & 1 & -y \\ 1 & 0 & -x \\ 0 & 0 & 0 \end{pmatrix} = -yX - xY + Z$$

Either making use of the decomposition into basis elements or by direct calculation, we have

$$gAg^{-1} = (a_1^0 \cosh \theta + a_2^0 \sinh \theta - a_3^0 y)X + (a_1^0 \sinh \theta + a_2^0 \cosh \theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0 \cosh \theta + b_2^0 \sinh \theta - b_3^0 y)X + (b_1^0 \sinh \theta + b_2^0 \cosh \theta - b_3^0 x)Y + b_3^0 Z$$

Indeed, the transformations here are combinations of hyperbolic rotations determined by θ , and translations determined by x and y .

The analysis requires two cases to be considered: when $B \in \text{Span}(X, Y)$ and when $B \notin \text{Span}(X, Y)$.

Case 1: $B \in \text{Span}(X, Y)$

Here, $b_3^0 = 0$ and $(b_1^0, b_2^0) \neq (0, 0)$, so we have

$$gAg^{-1} = (a_1^0 \cosh \theta + a_2^0 \sinh \theta - a_3^0 y)X + (a_1^0 \sinh \theta + a_2^0 \cosh \theta - a_3^0 x)Y + a_3^0 Z$$

$$gBg^{-1} = (b_1^0 \cosh \theta + b_2^0 \sinh \theta)X + (b_1^0 \sinh \theta + b_2^0 \cosh \theta)Y$$

If $b_1^0 \neq 0$ and $b_2^0 = 0$, then

$$\begin{aligned} (A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_1^0 X) \\ &\sim (a_2^0 Y + a_3^0 Z, b_1^0 X) \end{aligned}$$

Now, if $a_3^0 = 0$, then

$$\begin{aligned} (A, B) &\sim (a_2^0 Y, b_1^0 X) \\ &\sim (a_2^0 Y, a_2^0 X) \\ &\sim (Y, X) \end{aligned}$$

If $a_3^0 \neq 0$, then we apply a transformation consisting only of a shift in Y . This is done by choosing $\theta = 0$ so there is no rotation, $y = 0$ so there is no shift in X , and then taking $x = a_2^0/a_3^0$ to shift Y to zero. This is demonstrated as follows

$$\begin{aligned} (A, B) &\sim (g(a_2^0 Y + a_3^0 Z)g^{-1}, g(b_1^0 X)g^{-1}) \\ &= ((a_2^0 - a_3^0 x)Y + a_3^0 Z, b_1^0 X) \\ &= (a_3^0 Z, b_1^0 X) \\ &\sim (a_3^0 Z, a_3^0 X) \\ &\sim (Z, X) \end{aligned}$$

If $b_1^0 = 0$ and $b_2^0 \neq 0$, then

$$\begin{aligned} (A, B) &= (a_1^0 X + a_2^0 Y + a_3^0 Z, b_2^0 Y) \\ &\sim (a_1^0 X + a_3^0 Z, b_2^0 Y) \end{aligned}$$

If $a_3^0 = 0$, then

$$\begin{aligned} (A, B) &\sim (a_1^0 X, b_2^0 Y) \\ &\sim (a_1^0 X, a_1^0 Y) \\ &\sim (X, Y) \end{aligned}$$

If $a_3^0 \neq 0$, then we apply a transformation consisting only of a shift in X . This is done by choosing $\theta = 0$ so there is no rotation, $x = 0$ so there is no shift in Y , and then taking $y = a_1^0/a_3^0$ to shift X to zero.

$$\begin{aligned}
(A, B) &\sim (g(a_1^0 X + a_3^0 Z)g^{-1}, g(b_2^0 Y)g^{-1}) \\
&= ((a_1^0 - a_3^0 y)X + a_3^0 Z, b_2^0 Y) \\
&= (a_3^0 Z, b_2^0 Y) \\
&\sim (a_3^0 Z, a_3^0 Y) \\
&\sim (Z, Y)
\end{aligned}$$

The most general case is $b_1^0 \neq 0$ and $b_2^0 \neq 0$. The main idea here is that we will try to use hyperbolic rotations to simplify B . Referring to the form of B after a hyperbolic rotation, given by

$$gBg^{-1} = (b_1^0 \cosh \theta + b_2^0 \sinh \theta)X + (b_1^0 \sinh \theta + b_2^0 \cosh \theta)Y,$$

we see that if we want to reduce one of the coefficients to 0, we need to solve for θ in $\tanh \theta = -b_1^0/b_2^0$ or $-b_2^0/b_1^0$. $\tanh$ is a mapping $\mathbb{R} \mapsto (-1, 1)$. There is no solution when $b_1^0 = b_2^0$. So three different cases need to be analyzed, depending on the ratio of b_1^0 and b_2^0 .

If $b_1^0 = b_2^0$, then

$$\begin{aligned}
(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^0 X + b_1^0 Y)
\end{aligned}$$

In this case, if $a_3^0 = 0$, then

$$\begin{aligned}
(A, B) &\sim (a_1^1 X + a_2^1 Y, b_1^0 X + b_1^0 Y) \\
&\sim (a_1^2 X, b_1^0(X + Y)) \\
&\sim (a_1^2 X, a_1^2(X + Y)) \\
&\sim (X, X + Y)
\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, then take another $g \in SH(2)$ with $\theta = 0$ so that there is no rotation. Applying the transformation gives

$$\begin{aligned}
(A, B) &\sim (g(a_1^1 X + a_2^1 Y + a_3^0 Z)g^{-1}, g(b_1^0 X + b_1^0 Y)g^{-1}) \\
&\sim ((a_1^1 - a_3^0 y)X + (a_2^1 - a_3^0 x)Y + a_3^0 Z, b_1^0(X + Y))
\end{aligned}$$

It is possible to choose $x, y \in \mathbb{R}$ such that $a_1^1 - a_3^0 y = 0$ and $a_2^1 - a_3^0 x = 0$. Hence

$$\begin{aligned}
(A, B) &\sim (a_3^0 Z, b_1^0(X + Y)) \\
&\sim (a_3^0 Z, a_3^0(X + Y)) \\
&\sim (Z, X + Y)
\end{aligned}$$

If $b_1^0 < b_2^0$, then the equation $\tanh \theta = -b_1^0/b_2^0$ has a solution. There exists a $\theta \in \mathbb{R}$ such that $b_1^0 \cosh \theta + b_2^0 \sinh \theta = 0$, so $gBg^{-1} = b_2^1 Y$.

$$\begin{aligned}
(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_2^1 Y) \\
&\sim (a_1^1 X + a_3^0 Z, b_2^1 Y)
\end{aligned}$$

In this case, if $a_3^0 = 0$, then

$$\begin{aligned}
(A, B) &\sim (a_1^1 X, b_2^1 Y) \\
&\sim (a_1^1 X, a_1^1 Y) \\
&\sim (X, Y)
\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, use another $g \in SH(2)$

$$\begin{aligned}
(A, B) &\sim (g(a_1^1 X + a_3^0 Z)g^{-1}, g(b_2^1 Y)g^{-1}) \\
&= ((a_1^1 - a_3^0 y)X + a_3^0 Z, b_2^1 Y) \\
&= (a_3^0 Z, b_2^1 Y) \\
&\sim (a_3^0 Z, a_3^0 Y) \\
&\sim (Z, Y)
\end{aligned}$$

where I chose $\theta = 0$ so that there is no rotation, $x = 0$ so that Y is unchanged, and $y = a_1^1/a_3^0$ so that the coefficient $a_1^1 - a_3^0 y$ vanishes. This is the same idea as seen before. The two equivalent forms here have already been encountered.

If $b_2^0 < b_1^0$, then the equation $\tanh \theta = -b_2^0/b_1^0$ has a solution. There exists a $\theta \in \mathbb{R}$ such that $b_1^0 \sinh \theta + b_2^0 \cosh \theta = 0$, so $gBg^{-1} = b_1^1 X$.

$$\begin{aligned}
(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_1^1 X) \\
&\sim (a_2^1 Y + a_3^0 Z, b_1^1 X)
\end{aligned}$$

In this case, if $a_3^0 = 0$, then

$$\begin{aligned}
(A, B) &\sim (a_2^1 Y, b_1^1 X) \\
&\sim (a_2^1 Y, a_2^1 X) \\
&\sim (Y, X)
\end{aligned}$$

Otherwise, if $a_3^0 \neq 0$, use another $g \in SH(2)$

$$\begin{aligned}
(A, B) &\sim (g(a_2^1 Y + a_3^0 Z)g^{-1}, g(b_1^1 X)g^{-1}) \\
&= ((a_2^1 - a_3^0 x)Y + a_3^0 Z, b_1^1 X) \\
&= (a_3^0 Z, b_1^1 X) \\
&\sim (a_3^0 Z, a_3^0 X) \\
&\sim (Z, X)
\end{aligned}$$

where I chose $\theta = 0$ so that there is no rotation, $y = 0$ so that X is unchanged, and $x = a_2^1/a_3^0$ so that the coefficient $a_2^1 - a_3^0 x$ vanishes. Again, this is the same idea as seen before. The two equivalent forms here have already been encountered as well.

Case 2: $B \notin \text{Span}(X, Y)$

Here, $b_3^0 \neq 0$. Recalling the general form of gAg^{-1} and gBg^{-1} that was initially derived, it is possible to choose $x, y \in \mathbb{R}$ such that the coefficients $b_1^0 \cosh \theta + b_2^0 \sinh \theta - b_3^0 y = 0$ and $b_1^0 \sinh \theta + b_2^0 \cosh \theta - b_3^0 x = 0$, causing the X and Y components in B to vanish. We have

$$\begin{aligned}
(A, B) &\sim (gAg^{-1}, gBg^{-1}) \\
&= (a_1^1 X + a_2^1 Y + a_3^0 Z, b_3^0 Z) \\
&\sim (a_1^1 X + a_2^1 Y, b_3^0 Z)
\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 = 0$, then

$$\begin{aligned}
(A, B) &\sim (a_1^1 X, b_3^0 Z) \\
&\sim (a_1^1 X, a_1^1 Z) \\
&\sim (X, Z)
\end{aligned}$$

If $a_1^1 = 0$ and $a_2^1 \neq 0$, then

$$\begin{aligned}(A, B) &\sim (a_2^1 Y, b_3^0 Z) \\ &\sim (a_2^1 Y, a_2^1 Z) \\ &\sim (Y, Z)\end{aligned}$$

If $a_1^1 \neq 0$ and $a_2^1 \neq 0$, then just as before, there are three cases to analyze. If $a_1^1 = a_2^1$, then

$$\begin{aligned}(A, B) &\sim (a_1^1(X + Y), b_3^0 Z) \\ &\sim (a_1^1(X + Y), a_1^1 Z) \\ &\sim (X + Y, Z)\end{aligned}$$

Otherwise, in order to simplify further, we need to choose a $g \in SH(2)$ that is also an element of the stabiliser group $Stab(Z)$, where

$$Stab(Z) = \{g \in SH(2) : gZg^{-1} = Z\}$$

It is clear that we require $x = 0$ and $y = 0$. We have

$$\begin{aligned}(A, B) &\sim (g(a_1^1 X + a_2^1 Y)g^{-1}, g(b_3^0 Z)g^{-1}) \\ &= ((a_1^1 \cosh \theta + a_2^1 \sinh \theta)X + (a_1^1 \sinh \theta + a_2^1 \cosh \theta)Y, b_3^0 Z)\end{aligned}$$

We follow the same reasoning as before and try to apply rotations to simplify. If $a_1^1 < a_2^1$, then there exists a $\theta \in \mathbb{R}$ such that the coefficient $a_1^1 \cosh \theta + a_2^1 \sinh \theta = 0$. Hence

$$\begin{aligned}(A, B) &\sim (a_2^2 Y, b_3^0 Z) \\ &\sim (a_2^2 Y, a_2^2 Z) \\ &\sim (Y, Z)\end{aligned}$$

On the other hand, if $a_2^1 < a_1^1$, then there exists a $\theta \in \mathbb{R}$ such that the coefficient $a_1^1 \sinh \theta + a_2^1 \cosh \theta = 0$. Hence

$$\begin{aligned}(A, B) &\sim (a_1^2 X, b_3^0 Z) \\ &\sim (a_1^2 X, a_1^2 Z) \\ &\sim (X, Z)\end{aligned}$$

3.6 Summary

The following table summarizes the model control systems to which control systems over each Lie group are equivalent.

$SU(2)$	(u_1, u_3)
$SL(2)$	(E, H) (F, H) $(\pm E \pm F, H)$ $(E, E - F)$ (H, E) (F, E)
H	(X, Z) (Z, X) (Y, X)
$SE(2)$	(Y, X) (Z, X) (X, Z)
$SH(2)$	(X, Z) (Y, Z) $(X + Y, Z)$ $(X, X + Y)$ $(Z, X + Y)$ (Z, Y) (X, Y) (Z, X) (Y, X)

Table 1: Summary of Model Control Systems

4 Small-Time Controllability

In this section, we will determine the small time controllability of all the equivalent systems listed in Table 1.

In practice, the overall strategy to determine whether the pair (A, B) on the Lie algebra $\mathfrak{g}$ is small-time controllable or not is given as follows. First, we must compute the commutator $[A, B]$, and check if

$$\text{Span}\{A, B, [A, B]\} = \mathfrak{g}$$

That is, if the pair is bracket generating. We are working with Lie groups of dimension 3, so we want $[A, B]$ to be the third direction. Otherwise, if the pair is not bracket generating, then we can already conclude that the system is not controllable. If it is bracket generating, we can continue. Compute e^{tB} , and further define its closure $L = \overline{e^{tB}}$. In most of our cases, the closure is equal to the matrix exponential itself. Then, compute

$$\text{Ad}_L A = e^{tB} A e^{-tB}$$

Next, we determine the set

$$\text{conv}(\text{Ad}_L A) / \mathbb{R}B$$

This is to quotient out the “fast” direction, and study the convexification of what remains. Note that it should not matter whether we convexify first then quotient, or quotient first then convexify. If this set is bounded, then the system cannot be small-time controllable. If this set contains an infinite line, then we just need to check if the bracket $[\text{Ad}_L A, B]$ is still “fast”. This is done by checking if

$$[\text{Ad}_L A, B] \in \text{Span}\{\text{Ad}_L A, B\}$$

If this is true, then the system is small-time controllable. The most difficult cases are when the convexified set is semi-infinite. In these cases, we need to study the topology of the group more closely.

The control u can be taken to be very large. So in the system $\dot{g} = (A + uB)g$, we can neglect A . This puts us in a position to use the Rashevsky-Chow theorem, hence why we begin by checking the bracket generating condition.

4.1 Controllability on $SU(2)$

The system (u_1, u_3) is not small-time controllable, because the convexified set is a disc, bounded by a circle.

4.2 Controllability on $SL(2)$

4.2.1 (E, H)

Notice that $[E, H] = -2E$, and so the F direction does not appear. The system is not bracket generating, and so it cannot be small-time controllable. Another way to see this is that linear combinations of E and H generate upper-triangular matrices of the form

$$\begin{pmatrix} b & a \\ 0 & -b \end{pmatrix}$$

The constant 0 makes it clear that the system cannot be small-time controllable.

4.2.2 (F, H)

This case is similar to the previous one. The commutator $[F, H] = 2F$, and so the E direction does not appear. The system is not bracket generating, thus it cannot be small-time controllable. Another way to see this is that linear combinations of F and H generate lower-triangular matrices, from which it is obvious that the system cannot be small-time controllable.

4.2.3 $(E + F, H)$

Convexified set is the region above one branch of a hyperbola. We explicitly compute the solution to the system

$$\dot{g} = ((E + F) + uH)g = \begin{pmatrix} u & 1 \\ 1 & -u \end{pmatrix} g$$

The solution to this matrix differential equation is

$$g(t) = \exp \left(\begin{pmatrix} u & 1 \\ 1 & -u \end{pmatrix} t \right) g_0$$

where the matrix exponential is

$$\exp \left(\begin{pmatrix} u & 1 \\ 1 & -u \end{pmatrix} t \right) = \cosh(\sqrt{u^2 + 1}t)I + \frac{\sinh(\sqrt{u^2 + 1}t)}{\sqrt{u^2 + 1}}(E + F + uH)$$

for all u .

4.2.4 $(E, E - F)$

Convexified set is the region bounded by an ellipse. Hence the system is not small-time controllable.

4.2.5 (H, E)

Not bracket generating, hence not small-time controllable.

4.2.6 (F, E)

The convexified set, after we quotient out the “fast” direction E , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vE}}F)/\mathbb{R}E = \{F + vH : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vE}}F) + uE = \{F + uE + vH : u, v \in \mathbb{R}\}$$

Here, we have unlocked a new direction for the control, H . Now, the E and H directions can be made arbitrarily large by taking large u and v . This effectively makes F negligible. We check the bracket

$$[H, E] = 2E \in \text{span}\{H, E\}$$

E and H were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (F + uE + vH)g$$

Doing so is allowed because the commutator $[H, E]$ remains in the span of H and E , and thus the (closure of the) reachable set is the same as the original system. This new system has two fast directions, E and H , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uE+vH}}F)/\text{span}\{E, H\} = \{e^{-2v}F : v \in \mathbb{R}\}$$

This one-dimensional set is the positive half of an infinite line in the F direction, not including the origin.

Remark. When computing $Ad_{e^{uE+vH}}F$, we need the inverse matrix exponential, e^{-uE-vH} . Mathematically, this is possible for any matrix. However, possible restrictions on the controls mean we need to be more cautious. After the first quotient, the convexified set was an infinite straight line not passing through the origin in the F - H plane. By taking points on the line to the limit (taking large v), we can approximate an infinite line along the H axis passing the origin, that most importantly, goes in both directions. We were able to obtain both directions for free. Thanks to this, we can indeed directly use e^{-uE-vH} in this scenario. This is different from the hyperbola cases, where we only have the positive half of the line.

Let us now check if the solution to the system

$$\dot{g} = (wF + uE + vH)g, \quad u, w > 0$$

is periodic. If it is, then the periodicity allows this set to be viewed as an infinite line passing through the origin. The solution to this matrix differential equation is

4.3 Controllability on H

4.3.1 (X, Z)

Not bracket generating, hence not small-time controllable.

4.3.2 (Z, X)

Not bracket generating, hence not small-time controllable.

4.3.3 (Y, X)

The convexified set, after we quotient out the “fast” direction X , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vX}}Y)/\mathbb{R}X = \{Y + vZ : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vX}}Y) + uX = \{Y + uX + vZ : u, v \in \mathbb{R}\}$$

Now, the X and Z directions can be made arbitrarily large by taking large u and v . This effectively makes Y negligible. We check the bracket

$$[Z, X] = 0 \in \text{span}\{Z, X\}$$

X and Z were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Y + uX + vZ)g$$

Doing so is allowed because the commutator $[Z, X]$ remains in the span of Z and X , and thus the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Z , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uX+vZ}}Y)/\text{span}\{X, Z\}$$

We compute

$$e^{uX+vZ} = \exp \left(\begin{pmatrix} 0 & u & v \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & u & v \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uX+vZ}}Y = \begin{pmatrix} 1 & 0 & u \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \{Y + uZ : u \in \mathbb{R}\}$$

Convexifying this set,

$$\text{conv}(Ad_{e^{uX+vZ}}Y) = \{Y + uZ : u \in \mathbb{R}\}$$

We then quotient out the “fast” X and Z directions, giving

$$\text{conv}(Ad_{e^{uX+vZ}}Y)/\text{span}\{X, Z\} = \{Y\}$$

This one-dimensional set is bounded. This suggests the system is not small-time controllable.

4.4 Controllability on $SE(2)$

4.4.1 (Y, X)

Not bracket generating, hence not small-time controllable.

4.4.2 (Z, X)

The convexified set, after we quotient out the “fast” direction X , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vX}}Z)/\mathbb{R}X = \{Z - vY : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vX}}Z) + uX = \{Z + uX - vY : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily large by taking large u and v . This effectively makes Z negligible. We check the bracket

$$[Y, X] = 0 \in \text{span}\{Y, X\}$$

X and Y were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Z + uX - vY)g$$

Doing so is allowed because the commutator $[Y, X]$ remains in the span of Y and X , and thus the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Y , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\}$$

We compute

$$e^{uX-vY} = \exp \left(\begin{pmatrix} 0 & 0 & u \\ 0 & 0 & -v \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & 0 & u \\ 0 & 1 & -v \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uX-vY}}Z = \begin{pmatrix} 0 & -1 & -v \\ 1 & 0 & -u \\ 0 & 0 & 0 \end{pmatrix} = \{Z - vX - uY : u, v \in \mathbb{R}\}$$

Convexifying this set gives a plane passing through Z parallel to the X - Y plane. We then quotient out the “fast” X and Y directions, giving

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\} = \{Z\}$$

which is a one-dimensional bounded set. This suggests the system is not small-time controllable.

4.4.3 (X, Z)

Convexified set is the region bounded by a circle. Hence the system is not small-time controllable.

4.5 Controllability on $SH(2)$

4.5.1 (X, Z)

Convexified set is the region to the right of one branch of a hyperbola. We explicitly compute the solution to the system

$$\dot{g} = (X + uZ)g = \begin{pmatrix} 0 & u & 1 \\ u & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} g$$

The solution to this matrix differential equation is

$$g(t) = \exp \left(\begin{pmatrix} 0 & u & 1 \\ u & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} t \right) g_0$$

where the matrix exponential is

$$\exp \left(\begin{pmatrix} 0 & u & 1 \\ u & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} t \right) = \begin{pmatrix} \cosh ut & \sinh ut & \frac{1}{u} \sinh ut \\ \sinh ut & \cosh ut & \frac{1}{u} (\cosh ut - 1) \\ 0 & 0 & 1 \end{pmatrix}$$

4.5.2 (Y, Z)

Convexified set is the region above one branch of a hyperbola. We explicitly compute the solution to the system

$$\dot{g} = (Y + uZ)g = \begin{pmatrix} 0 & u & 0 \\ u & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} g$$

The solution to this matrix differential equation is

$$g(t) = \exp \left(\begin{pmatrix} 0 & u & 0 \\ u & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} t \right) g_0$$

where the matrix exponential is

$$\exp \left(\begin{pmatrix} 0 & u & 0 \\ u & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} t \right) = \begin{pmatrix} \cosh ut & \sinh ut & \frac{1}{u} (\cosh ut - 1) \\ \sinh ut & \cosh ut & \frac{1}{u} \sinh ut \\ 0 & 0 & 1 \end{pmatrix}$$

4.5.3 $(X + Y, Z)$

Not bracket generating, hence not small-time controllable.

4.5.4 $(X, X + Y)$

Not bracket generating, hence not small-time controllable.

4.5.5 $(Z, X + Y)$

Not bracket generating, hence not small-time controllable.

4.5.6 (Z, Y)

The convexified set, after we quotient out the “fast” direction Y , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vY}}Z)/\mathbb{R}Y = \{Z - vX : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vY}}Z) + uY = \{Z + uY - vX : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily large by taking large u and v . This effectively makes Z negligible. We check the bracket

$$[X, Y] = 0 \in \text{span}\{X, Y\}$$

X and Y were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Z + uY - vX)g$$

Doing so is allowed because the commutator $[X, Y]$ remains in the span of X and Y , and thus the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Y , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uY-vX}}Z)/\text{span}\{X, Y\}$$

We compute

$$e^{uY-vX} = \exp \left(\begin{pmatrix} 0 & 0 & -v \\ 0 & 0 & u \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & 0 & -v \\ 0 & 1 & u \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uY-vX}}Z = \begin{pmatrix} 0 & 1 & -u \\ 1 & 0 & v \\ 0 & 0 & 0 \end{pmatrix} = \{Z - uX + vY : u, v \in \mathbb{R}\}$$

Convexifying this set gives a plane passing through Z parallel to the X - Y plane. We then quotient out the “fast” X and Y directions, giving

$$\text{conv}(Ad_{e^{uY-vX}}Z)/\text{span}\{X, Y\} = \{Z\}$$

which is a one-dimensional bounded set. This suggests the system is not small-time controllable.

4.5.7 (X, Y)

Not bracket generating, hence not small-time controllable.

4.5.8 (Z, X)

The convexified set, after we quotient out the “fast” direction X , is a straight line not passing through the origin. It is given by

$$\text{conv}(Ad_{e^{vX}}Z)/\mathbb{R}X = \{Z - vY : v \in \mathbb{R}\}$$

Taking the “fast” direction into account again as in the original system, we have

$$\text{conv}(Ad_{e^{vX}}Z) + uX = \{Z + uX - vY : u, v \in \mathbb{R}\}$$

Now, the X and Y directions can be made arbitrarily large by taking large u and v . This effectively makes Z negligible. We check the bracket

$$[Y, X] = 0 \in \text{span}\{Y, X\}$$

X and Y were unable to generate a new direction. We need to continue to find quotient spaces, which gives us new directions to use. Now we consider a modified system

$$\dot{g} = (Z + uX - vY)g$$

Doing so is allowed because the commutator $[X, Y]$ remains in the span of X and Y , and thus the (closure of the) reachable set is the same as the original system. This new system has two fast directions, X and Y , which we quotient from the convexified Ad orbit. This is done by computing

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\}$$

We compute

$$e^{uX-vY} = \exp \left(\begin{pmatrix} 0 & 0 & u \\ 0 & 0 & -v \\ 0 & 0 & 0 \end{pmatrix} \right) = \left\{ \begin{pmatrix} 1 & 0 & u \\ 0 & 1 & -v \\ 0 & 0 & 1 \end{pmatrix} : u, v \in \mathbb{R} \right\}$$

So

$$Ad_{e^{uX-vY}}Z = \begin{pmatrix} 0 & 1 & v \\ 1 & 0 & -u \\ 0 & 0 & 0 \end{pmatrix} = \{Z + vX - uY : u, v \in \mathbb{R}\}$$

Convexifying this set gives a plane passing through Z parallel to the X - Y plane. We then quotient out the “fast” X and Y directions, giving

$$\text{conv}(Ad_{e^{uX-vY}}Z)/\text{span}\{X, Y\} = \{Z\}$$

which is a one-dimensional bounded set. This suggests the system is not small-time controllable.

4.5.9 (Y, X)

Not bracket generating, hence not small-time controllable.

4.6 Summary

The following table summarizes whether each of the model systems found is small-time controllable or not.

$SU(2)$	(u_1, u_3)	No
$SL(2)$	(E, H)	No
	(F, H)	No
	$(\pm E \pm F, H)$	?
	$(E, E - F)$	No
	(H, E)	No
	(F, E)	?
H	(X, Z)	No
	(Z, X)	No
	(Y, X)	?
$SE(2)$	(Y, X)	No
	(Z, X)	?
	(X, Z)	No
$SH(2)$	(X, Z)	?
	(Y, Z)	?
	$(X + Y, Z)$	No
	$(X, X + Y)$	No
	$(Z, X + Y)$	No
	(Z, Y)	?
	(X, Y)	No
	(Z, X)	?
	(Y, X)	No

Table 2: Summary of Small-Time Controllability

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